

100%

Name: _____

These questions assess your understanding of the material from Chapter(s) 16, 17 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Suppose a train sounding its 306-Hz horn is moving at 38.9 $\frac{m}{s}$. What frequency does a person standing by the tracks hear as the train approaches? Given this particular day's temperature, the speed of sound is 339 $\frac{m}{s}$.

ANSWER: 346 Hz

$$V_w = 339 \frac{m}{s} \quad f_{obs} = f_s \left(\frac{V_w}{V_w - V_s} \right)$$

$$f = 306 \text{ Hz} \quad = 306 \text{ Hz} \left(\frac{339 \frac{m}{s}}{339 - 38.9} \right)$$

$$V_s = 38.9 \frac{m}{s}$$

$$f_{obs} = ? \quad = 306 (1.1296) = 345.66 \text{ Hz}$$

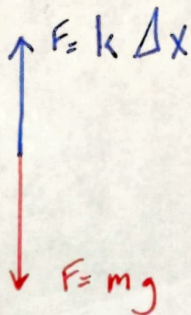
2. When Adam gets in his car, it lowers by 0.62 cm. If Adam's mass is 104 kg, then what is the force constant of his car's suspension system?

ANSWER: $1.64 \times 10^5 \frac{N}{m}$

$$x = 0.62 \times 10^{-2} \text{ m}$$

$$m = 104 \text{ kg}$$

$$a_g = 9.80 \frac{m}{s^2}$$



$$\frac{k(0.62 \times 10^{-2})}{0.62 \times 10^{-2}} = \frac{104(9.8)}{0.62 \times 10^{-2}}$$

$$k = 1.6439 \times 10^5$$

3. Ultrasound is transmitted from fat to muscle during a particular medical procedure. Calculate the percentage of the ultrasound's intensity that is reflected when the ultrasound travels from fat with a density of 979 $\frac{kg}{m^3}$ to muscle tissue with a density of $1.115 \times 10^3 \frac{kg}{m^3}$. The speed of ultrasound in the fat is $1.38 \times 10^3 \frac{m}{s}$ and the speed of ultrasound in the muscle tissue is $1.57 \times 10^3 \frac{m}{s}$.

ANSWER: 1.67 %

$$\rho_f = 979 \frac{kg}{m^3}$$

$$Z = \rho v$$

$$Z_{1,f} = 979 \frac{kg}{m^3} \left(1.38 \times 10^3 \frac{m}{s} \right) = 1.351 \times 10^6 \frac{kg}{m^2 \cdot s}$$

$$\rho_m = 1.115 \times 10^3 \frac{kg}{m^3}$$

$$Z_{2,m} = 1.115 \times 10^3 (1.57 \times 10^3) = 1.7506 \times 10^6$$

$$V_f = 1.38 \times 10^3 \frac{m}{s}$$

$$V_m = 1.57 \times 10^3 \frac{m}{s}$$

$$a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2} = \frac{(1.7506 \times 10^6 - 1.351 \times 10^6)^2}{(1.351 \times 10^6 + 1.7506 \times 10^6)^2} = 0.016544 \Rightarrow 1.6544 \%$$

+3

4. Suppose that a 2.20×10^3 kg car has a suspension system with a force constant $3.94 \times 10^4 \frac{\text{N}}{\text{m}}$. If the car hits a bump and bounces with an amplitude of 13. cm, then what is its maximum vertical velocity assuming no damping occurs?

ANSWER: 0.55 m/s

$$M = 2.20 \times 10^3 \text{ kg}$$

$$k = 3.94 \times 10^4 \frac{\text{N}}{\text{m}}$$

$$X = 13 \times 10^{-2} \text{ m}$$

$$V = \sqrt{\frac{k}{m}} X$$

$$= \sqrt{\frac{3.94 \times 10^4}{2.2 \times 10^3}} (0.13)$$

$$= 0.55015 \frac{\text{m}}{\text{s}}$$

5. Calculate the sound intensity level in decibels for a sound wave traveling in air at 0.25°C with a pressure amplitude of 0.0688 Pa. Recall that the density of air is $1.29 \frac{\text{kg}}{\text{m}^3}$ and the speed of sound in air at 0°C is $331 \frac{\text{m}}{\text{s}}$.

ANSWER: 67.4 dB

$$T = 0.25 + 273.15 = 273.40^\circ\text{K}$$

$$\rho_{\text{air}} = 1.29 \quad V_w = 331 \frac{\text{m}}{\text{s}} \sqrt{\frac{T}{273}}$$

$$P = 0.0688 \text{ Pa}$$

$$= 331 \sqrt{\frac{273.40}{273}}$$

$$= 331.25 \frac{\text{m}}{\text{s}}$$

$$I = \frac{(\Delta P)^2}{2 \rho V_w} \quad B_{\text{dB}} = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

$$= \frac{(0.0688)^2}{2(1.29)(331.25)} = \frac{10 \log_{10} \left(\frac{5.5385 \times 10^{-6}}{1 \times 10^{-12}} \right)}{854.63}$$

$$= \frac{4.7334 \times 10^{-3}}{854.63} = 5.5385 \times 10^{-6}$$

6. A cello generates a note with a pitch of 665. Hz. What is the period of the string's oscillation?

ANSWER: $1.5 \times 10^{-3} \text{ sec}$

$$f_c = 665 \text{ Hz}$$

$$T = \frac{1}{f}$$

$$= \frac{1}{665 \text{ Hz}}$$

$$= 1.5038 \times 10^{-3}$$

7. Suppose you have a tube that is closed at one end. If you want its fundamental frequency to be 359 Hz and the air temperature is 0.86°C, then what should the length of the tube be?

ANSWER: 0.23 m

$$n = 1$$

$$f_n = 359 \text{ Hz}$$

$$T = 0.86 + 273.15 = 2.7401 \times 10^2 = 274.01$$

$$v_w = 331 \sqrt{\frac{T}{273 \text{ K}}}$$

$$= 331 \sqrt{\frac{274.01}{273}} = 331.61 \frac{\text{m}}{\text{s}}$$

$$f_n = n \frac{v_w}{4L}$$

$$359 \text{ Hz} = 1 \left(\frac{331.61}{4L} \right)$$

$$\frac{1.436 \times 10^3 L}{1.436 \times 10^3} = \frac{331.61}{1.436 \times 10^3}$$

$$L = 0.23093 \text{ m}$$

✓

8. The spring of a toy gun has a force constant of 55.9 N/m and is compressed 25.5 cm. Neglecting friction and the mass of the spring, at what speed will a 1.7-g projectile be ejected from the toy gun?

ANSWER: 46 m/s

$$k = 55.9 \frac{\text{N}}{\text{m}}$$

$$m = 1.7 \times 10^{-3} \text{ kg}$$

$$x = 25.5 \times 10^{-2} = 0.255 \text{ m}$$

$$v = \sqrt{\frac{k}{m}} x$$

$$= \sqrt{\frac{55.9}{1.7 \times 10^{-3}}} x$$

$$= \sqrt{3.2882 \times 10^4} x$$

$$= (1.8133 \times 10^2) (0.255)$$

$$= 46.239$$

✓

9. What is the length of a pendulum that has a period of 0.49 s?

ANSWER: 0.0596 m

$$T = 0.49 \text{ s}$$

$$T_{\text{pend}} = 2\pi \sqrt{\frac{L}{g}}$$

$$\frac{0.49 \text{ s}}{2\pi} = \sqrt{\frac{L}{g}}$$

$$0.077986 = \sqrt{\frac{L}{g}}$$

$$6.0818 \times 10^{-3} = \frac{L}{g} \Rightarrow 5.96 \times 10^{-2} = 0.0596 \text{ m}$$

+3

PHY202 Exam 1 on chapter(s): 16, 17

1. $f = 345.7 \text{ Hz} \therefore f_{obs} = f_s \left(\frac{v_w}{v_w \pm v_s} \right)$ (review the Doppler effect)
(Chapter 17, Section (4) Doppler Effect and Sonic Booms - cp1e_ch17_ex4)
2. $k = 1.64 \times 10^5 \frac{\text{N}}{\text{m}} \therefore \vec{F} = -k\Delta\vec{x}$. $k = -\frac{\vec{F}}{\Delta\vec{x}}$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex1)
3. $a = 1.66\% \therefore Z = \rho v$, $a = \frac{(z_2 - z_1)^2}{(z_1 + z_2)^2}$
(Chapter 17, Section (7) Ultrasound - cp1e_ch17_ex7)
4. $v_{max} = 0.550 \frac{\text{m}}{\text{s}} \therefore v(t) = \sqrt{\frac{k}{m}} X \sin \frac{2\pi t}{T}$ and the max of sine's range is 1.
(Chapter 16, Section (5) Energy and the Simple Harmonic Oscillator - cp1e_ch16_ex6)
5. $\beta = 67.4 \text{ dB} \therefore v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$, $I = \frac{(\Delta p)^2}{2\rho v_w}$, $\beta \text{ (dB)} = 10 \log_{10} \left(\frac{I}{I_0} \right)$, $I_0 = 1 \times 10^{-12} \frac{\text{W}}{\text{m}^2}$
(Chapter 17, Section (3) Sound Intensity and Sound Level - cp1e_ch17_ex2)
6. $T = 1.50 \text{ ms} \therefore T = \frac{1}{f}$
(Chapter 16, Section (2) Period and Frequency in Oscillations - cp1e_ch16_ex3)
7. $L = 0.2309 \text{ m} \therefore f_n = n \frac{v_w}{4L}$, $v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5a)
8. $v = 46.2 \frac{\text{m}}{\text{s}} \therefore KE = PE_{elastic}$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex2)
9. $L = 5.96 \text{ cm} \therefore T_{pend} = 2\pi \sqrt{\frac{L}{g}}$
(Chapter 16, Section (4) Simple Pendulum - cp1e_ch16_p22)